An Empirical Evaluation of Source Code Chunking and Context Headers for Requirement Evidence Retrieval

Anonymous Authors

***Abstract*—Locating source code that evidences a technical requirement is a common need in software analysis. Syntax-aware chunking is motivated by its preservation of code structure, yet chunk boundaries are confounded with the context metadata prepended to each chunk. In this Early Research Achievements (ERA) paper, we cross two chunking strategies, node-level abstract syntax tree (AST) chunking and 50-line windows, with and without a context header. We evaluate them on 25 job descriptions, whose required skills serve as requirements, and 40 public repositories (434 line-window and 438 AST chunks), using BM25, dense retrieval, and cross-encoder reranking against 882 human-labeled pairs. After reranking, headers improved normalized discounted cumulative gain at rank 10 (NDCG@10) for AST chunks from 0.341 to 0.487 ($p_{\text{Holm}} < 0.001$), and without headers, line windows outperformed AST chunks (0.580 vs. 0.341, $p_{\text{Holm}} < 0.001$). Line windows without headers significantly outperformed AST chunks with headers ($\Delta = -0.093$, $p_{\text{Holm}} = 0.0438$). Line windows with headers achieved the highest NDCG@10 (0.621), significantly above AST with headers in an exploratory comparison ($\Delta = -0.134$, unadjusted $p = 0.0003$). With the final parser configuration, the AST arm triggers no line fallback (0.0%), and an audit of boundary effects, coverage differences, and window overlap does not change the ranking of conditions; we close with a research roadmap for code retrieval.**

***Index Terms*—Source code retrieval, code chunking, abstract syntax trees, context headers, requirements traceability, empirical software engineering**

I. Introduction

Locating source code that provides concrete evidence for a technical requirement is fundamental in software analysis and capability assessment. While classical requirements traceability links formal specifications to code artifacts [1], [2], modern software engineering frequently demands verifying whether a codebase implements higher-level framework capabilities (e.g., reactive state management or token authentication). We formalize this task as **Requirement & Technical Capability Evidence Retrieval**: given natural-language technical requirements—instantiated via mandatory skill specifications from job descriptions (JDs)—the objective is to retrieve the specific code chunks that directly evidence their concrete implementation across multi-file repositories, without requiring explicit requirement IDs.

In retrieval-augmented generation (RAG) pipelines for software engineering, source code must be segmented into chunks for neural encoding. Chunking via abstract syntax trees (ASTs) is widely advocated to preserve semantic units (methods, declarations) [3]. However, recent empirical studies on code completion indicate that structure-aware chunking does not reliably outperform simple sliding windows [4], and line-based segmentations match syntax-aware splitting across compute budgets [5]. Crucially, prior work conflates chunk boundaries with the contextual metadata (file path, enclosing class, method signature) prepended to chunks. Whether structural chunking provides distinct advantages once metadata is controlled remains unexamined for requirement evidence retrieval.

In this Early Research Achievements (ERA) paper, we present a controlled 2×2 factorial evaluation crossing chunking strategy (node-level AST vs. 50-line windows) with context headers (present vs. absent). We address four research questions: **RQ1:** Does prepending a context header improve evidence retrieval from code chunks? **RQ2:** Does node-level AST chunking outperform line windows, with and without a context header? **RQ3 (Roadmap):** Which header components (file path, class, symbol, call-graph or import context) drive retrieval gains, and how do the trade-offs generalize across languages? **RQ4 (Roadmap):** Does retrieved evidence improve human accuracy and confidence in requirement verification?

We evaluate RQ1 and RQ2 across 25 JDs and 40 open-source repositories (20 Java Spring Boot, 20 React/TypeScript; 434 line-window and 438 AST chunks) at three tiers: BM25, dense bi-encoder (BGE-M3), and cross-encoder reranking (BGE-Reranker-Base), evaluated against 882 blinded human-labeled pairs ($\kappa_w = 0.9822$). Findings reveal: (1) headers significantly improve AST chunks ($\Delta = +0.146$, $p_{\text{Holm}} < 0.001$), but not line windows ($\Delta = +0.041$, $p_{\text{Holm}} = 0.2411$); (2) without headers, line windows significantly outperform AST chunks ($\Delta = -0.239$, $p_{\text{Holm}} < 0.001$); (3) headerless line windows significantly outperform AST chunks with headers ($\Delta = -0.093$, $p_{\text{Holm}} = 0.0438$); (4) line windows with headers achieve highest overall effectiveness (NDCG@10 $= 0.621$), outperforming AST with headers ($\Delta = -0.134$, unadjusted $p = 0.0003$); and (5) with the final parser configuration, the AST fallback rate is 0.0%, showing that the performance gap is not caused by fallback degeneration.

Contributions: (1) *Factorial Framework:* Isolating chunking boundaries from context headers in requirement evidence retrieval. (2) *Empirical Findings:* Evidence that sliding windows match or exceed node-level AST chunking, with headers substantially improving AST chunks (the underlying mechanism is untested; Section V). (3) *Curated Benchmark:* A replication package with 882 blinded dual-annotated pairs ($\kappa_w = 0.9822$, 100% Judged@5, zero orphaned pairs). (4) *ERA Roadmap:* An actionable research agenda addressing header components,

dependency graphs, and human practitioner verification.

II. Related Work

We organize related work into four domains:

Code Chunking for RAG: Line-based chunking can disrupt semantic boundaries [3]. cAST [3] addresses this by recursively partitioning AST nodes and merging siblings under size limits, reporting gains on RepoEval and SWE-bench. However, subsequent controlled experiments in code completion show that structure-aware chunking does not reliably outperform sliding windows [4], and line-based slicing matches syntax-aware splitting across compute budgets [5]. While those studies target code synthesis, we evaluate human-labeled requirement-to-code evidence retrieval and orthogonally control context headers.

Dense Retrieval and Reranking: Neural code search embeds natural-language queries and source code into shared representations [6], [7], [8]. Modern dense models such as BGE-M3 [9] perform multilingual semantic matching. To overcome single-vector compression limits, two-stage pipelines integrate cross-encoder rerankers that jointly attend across query and chunk tokens [10], [11]. We adopt this recall-then-rerank architecture to investigate how chunk geometry and headers interact with cross-encoders.

Context Enrichment: Isolated code chunks frequently lack enclosing class and package context. Prior approaches enrich completion prompts with repository context iteratively (RepoCoder [12]) or benchmark cross-file dependencies (RepoBench [13], CrossCodeEval [14]). We investigate a lightweight alternative: prepending standardized context headers to individual chunks prior to retrieval.

Requirements Traceability and Skill Mining: Tracing natural-language requirements to source code has deep roots in software engineering [1], [2]. Separately, repository mining has been employed to classify developer expertise from framework usage [15]. We adopt the traceability perspective: treating required technical skills as requirements and retrieving concrete code evidence, without assessing individual developers.

III. Approach and Study Design

To separate chunk boundaries from contextual metadata, we employ a 2×2 factorial design evaluated across three retrieval tiers, governed by a frozen analysis protocol.

A. Corpus, Candidate Space, and Curation

Our corpus comprises 40 open-source GitHub repositories (20 Java Spring Boot, 20 React/TypeScript) spanning 189 source files. As documented in `sampling_frame_log.json`, repositories were curated from public GitHub projects exhibiting production architectural patterns (controllers, services, DTOs, custom hooks, Redux stores, UI components). To focus evaluation on authentic capability evidence and avoid diluting the corpus with build artifacts, third-party libraries, test mocks, or auto-generated configuration, curation targeted the core functional architecture modules (mean 4.7 files per repository, 189 files total) where business logic, architectural patterns, and framework capabilities reside. Repository licenses encompass 19 MIT (47.5%), 16 Public GitHub default copyright (40.0%), 4 Apache-2.0 (10.0%), and 1 GNU AGPL-3.0 (2.5%); all source text is redistributed in our replication package under original author copyright. Queries represent 25 public job descriptions (15 Java backend, 10 React/TypeScript frontend) with mandatory technical skill requirements extracted via a frozen prompt on `gemini-1.5-flash` (`queries_frozen.json`). For each query, retrieval evaluates across the full candidate space of all 40 repositories (both languages, without language filtering): 434 chunks for line conditions and 438 chunks for AST conditions (872 chunk instances across the two strategies).

B. Factorial Conditions and Retrieval Pipeline

We evaluate two orthogonal dimensions: **Chunking Strategy:** *Line-window chunking* slices files into 50 lines of code (LOC) windows with 10-LOC overlap (step 40 LOC, minimum 6 LOC; 434 chunks, mean 41.0 LOC, median 50 LOC, 100% line coverage across all 189 files, 28.6% exceeding 512 tokens, 56.0% cutting syntactic boundaries). *AST chunking* extracts node-level units using Tree-sitter[1] parsers expanded to modern constructs: 438 chunks (134 methods/constructors, 96 arrow functions, 86 component wrappers, 62 functions, 28 class/interface/record/enum declarations, and 32 type/constant/config declarations; mean 26.7 LOC, median 16 LOC, 15.8% exceeding 512 tokens). AST chunks can be nested (e.g., a class declaration and its methods), so per-chunk LOC sums exceed the number of covered lines. Due to $6 \leq \text{LOC} \leq 85$ filtering, AST chunks cover 69.0% of lines (10,585/15,351) across 183/189 files (6 files contain zero AST chunks), maintaining 87.2% outer-boundary alignment (56 module-wrapped functions classified as cut by interval bounds, though representing intact AST nodes). No fallback was triggered (0/438 AST chunks, 0/250 top-10 candidates). The six files without any extractable unit (6/189) contribute no AST chunks and are therefore absent from the AST candidate space; the common-support analysis (Section IV-C) addresses this coverage difference. **Context Header Condition:** *Without Header* contains raw code. For *With Header*, line chunks prepend File and Lines cues (100%); AST chunks prepend File (100%), enclosing Class (35.6%), Method (100%), and Lines (100%) (`// File: <path> \n [// Class: <name> \n] // Method: <name> \n // Lines: <start>-<end>`). The Method field holds the name of the extracted node (method, function, component, type, or constant); Class is present only for nodes nested in a class (35.6% of AST chunks). **Retrieval Tiers:** (1) *BM25* [16] (lexical baseline). (2) *Dense* BAAI/bge-m3 [9] (1024d embeddings, cosine similarity). (3) *Reranking* BAAI/bge-reranker-base [17] (cross-encoder rescoring top-20 dense candidates).

C. Unified Blinded Pooling Protocol

To prevent system bias, we executed a single-wave Cranfield/TREC pooling protocol: top-5 candidates from all 12

[1] https://tree-sitter.github.io/

systems (BM25, Dense, Rerank $\times$ 4 conditions) were pooled, yielding 1,500 candidates deduplicated to 882 unique pairs. Two software engineering (SE) annotators independently labeled all 882 pairs on a pre-specified 3-point relevance rubric: **0** (irrelevant/boilerplate: imports, standard configuration, or code lacking implementation of the target capability), **1** (partially relevant/indirect: interface signatures, type definitions, or peripheral helpers), and **2** (direct evidence: concrete, executable implementation directly demonstrating the requested technical capability). Annotators were blinded to system, header condition, rank, and score (chunk geometry cannot be concealed, making strategy blinding partial). Inter-annotator agreement reached almost perfect concordance: quadratic-weighted Cohen's $\kappa = 0.9822$ (linear $\kappa = 0.9764$, unweighted $\kappa = 0.9708$, 98.07% exact agreement). The 17 disagreements were adjudicated by a Lead Adjudicator using standardized rubrics, yielding complete coverage (Judged@3 $= 100.0\%$, Judged@5 $= 100.0\%$) and zero orphaned pairs.

D. Evaluation Metrics and Hypothesis Testing

The primary metric is NDCG@10 [18] (linear gain $\mathrm{rel}_i / \log_2(i+1)$). Unjudged items receive relevance 0; we also report binary preference (`bpref`) [19], NDCG@1, mean reciprocal rank (MRR@10), and Judged@10. Judged@k is the fraction of top-k retrieved chunks carrying a human label. The pre-specified confirmatory family (fixed in an analysis plan before the final evaluation run and included in the replication package) tests four paired comparisons over $N = 25$ queries at the rerank tier: **Comp 1** (AST+Header vs. Line No-Header), **Comp 2** (AST No-Header vs. Line No-Header), **Comp 3** (AST+Header vs. AST No-Header), and **Comp 4** (Line+Header vs. Line No-Header) via two-sided Wilcoxon signed-rank tests [20] with Holm-Bonferroni correction [21] ($\alpha = 0.05$). We report paired effect sizes Δ with 95% bootstrap CIs (10,000 resamples), plus exploratory contrast **Comp 5** (AST+Header vs. Line+Header). Wilcoxon p-values are exact where no ties occur and use the normal approximation otherwise.

IV. Empirical Results

A. Overall Benchmark Results across Twelve Systems

Table I presents the 12-configuration benchmark across lexical (BM25), dense bi-encoder (BGE-M3), and cross-encoder reranking tiers. At reranking, Line+Header achieves the highest NDCG@10 (0.621, 95% CI $[0.555, 0.682]$). It ties Line No-Header on NDCG@1 (0.800), while Line No-Header has a marginally higher MRR@10 (0.953 vs. 0.950). AST+Header reaches 0.487 ($[0.419, 0.551]$), and AST No-Header drops to 0.341 ($[0.263, 0.421]$).

Dense encoding improves NDCG@10 over BM25 by 0.118–0.139 with headers, and reranking adds another 0.037–0.070. Header gains differ across chunking strategies. For AST chunks, they grow with retriever capacity: negligible for BM25 ($+0.006$), moderate for dense retrieval ($+0.092$), and large for reranking ($+0.146$). For line windows, gains are modest at every stage ($+0.028$, $+0.030$, $+0.041$). Because pooling spanned all 12 systems at top-5, Judged@5 is 100.0% for all configurations. Binary preference (`bpref`) orders the four conditions identically to NDCG@10 at every tier (Table I). At the reranking tier: Line+H 0.353 $>$ Line No-H 0.300 $>$ AST+H 0.241 $>$ AST No-H 0.170. This indicates that rankings are robust to unjudged items at deeper ranks. Two patterns deserve mention: at the dense tier AST+Header attains higher NDCG@1 (0.620 vs. 0.580) and MRR@10 (0.910 vs. 0.883) than Line+Header, although it trails on NDCG@10 (0.450 vs. 0.551); reranking improves every condition over dense except AST No-Header, dropping from 0.358 to 0.341, consistent with cross-encoder reliance on header cues for isolated AST chunks.

TABLE I
Retrieval effectiveness across 12 configurations ($N = 25$ JDs).

Tier	Chunk	Header	NDCG@10 [95% CI]	NDCG@1	MRR@10	bpref	J@10
Two-Stage Re-ranking Tier (Cross-Encoder)							
Rerank	Line	Yes	**0.621** [0.555, 0.682]	**0.800**	0.950	**0.353**	0.736
Rerank	Line	No	*0.580* [0.505, 0.649]	**0.800**	**0.953**	*0.300*	0.716
Rerank	AST	Yes	0.487 [0.419, 0.551]	0.740	0.896	0.241	0.660
Rerank	AST	No	0.341 [0.263, 0.421]	0.340	0.699	0.170	0.712
Dense Bi-Encoder Tier (BGE-M3)							
Dense	Line	Yes	**0.551** [0.480, 0.619]	0.580	0.883	**0.347**	0.752
Dense	Line	No	0.521 [0.439, 0.600]	0.600	0.891	0.297	0.720
Dense	AST	Yes	0.450 [0.393, 0.506]	**0.620**	**0.910**	0.247	0.692
Dense	AST	No	0.358 [0.298, 0.420]	0.420	0.813	0.179	0.716
Lexical Sparse Baseline Tier (BM25)							
BM25	Line	Yes	**0.433** [0.376, 0.486]	**0.440**	**0.868**	**0.255**	0.696
BM25	Line	No	0.405 [0.350, 0.458]	**0.440**	0.865	0.246	0.644
BM25	AST	Yes	0.311 [0.243, 0.376]	0.340	0.761	0.137	0.676
BM25	AST	No	0.305 [0.236, 0.371]	0.340	0.724	0.129	0.560

Bold: best within tier; italic: second best (Rerank tier). Linear gain $\mathrm{rel}/\log_2(i+1)$. J@10: Judged@10 coverage.

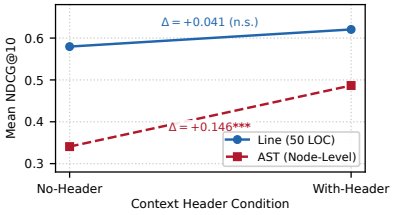


Fig. 1. Interaction plot of context header sensitivity at reranking ($N = 25$).

B. Hypothesis Testing and RQ Analyses

Table II reports the pre-specified confirmatory Wilcoxon signed-rank tests with Holm-Bonferroni correction, alongside the exploratory contrast.

RQ1 (Context Headers): Prepending headers yields a large, significant gain for AST chunks (**Comp 3**: $\Delta = +0.146$, 95% CI $[+0.086, +0.207]$, $p_{\mathrm{Holm}} < 0.001$, $W = 24.0$; AST+H wins 23/25 JDs). In contrast, prepending headers to line windows yields a minor, non-significant gain (**Comp 4**: $\Delta = +0.041$, 95% CI $[-0.012, +0.093]$, $p_{\mathrm{Holm}} = 0.2411$, $W = 118.0$; 15 wins vs. 10 losses). A post hoc exploratory difference-in-differences test (unadjusted) indicates that the header effect differs between strategies ($W = 66.0, p = 0.0081$). As shown in Fig. 1, headers significantly improve AST chunks, whereas the gain for 50-line windows is not statistically significant.

TABLE II
HYPOTHESIS TESTS ON NDCG@10 AT RERANKING TIER ($N = 25$).

ID	Contrast ($A - B$)	Δ [95% CI]	W/L/T	p_{Holm}
Confirmatory family (step-down Holm, $\alpha = 0.05$)				
Comp 1	AST+Header vs. Line No-Header	-0.093 $[-0.164, -0.023]$	7/18/0	0.0438^{*}
Comp 2	AST No-Header vs. Line No-Header	-0.239 $[-0.292, -0.185]$	2/23/0	$< 0.001^{***}$
Comp 3	AST+Header vs. AST No-Header	$+0.146$ $[+0.086, +0.207]$	23/2/0	$< 0.001^{***}$
Comp 4	Line+Header vs. Line No-Header	$+0.041$ $[-0.012, +0.093]$	15/10/0	0.2411
Exploratory (outside the Holm family)				
Comp 5	AST+Header vs. Line+Header	-0.134 $[-0.199, -0.070]$	5/20/0	$< 0.001^{\dagger}$

W/L/T: queries where A wins, loses, or ties against B. Significance follows p_{Holm} ($^{*} < 0.05$, $^{**} < 0.01$, $^{***} < 0.001$). †Unadjusted p_{raw}.

RQ2 (AST vs. Line): Without headers, RQ2 is tested by Comp 2 (confirmatory); with headers, by Comp 5 (exploratory). Comp 1 additionally compares across header conditions (AST+Header vs. Line No-Header). Simple line windows without headers significantly outperform AST chunks with headers (**Comp 1**: $\Delta = -0.093$, 95% CI $[-0.164, -0.023]$, $p_{\text{Holm}} = 0.0438$, $W = 78.0$; Line wins 18 vs. AST wins 7). Without headers, line windows strongly outperform AST chunks (**Comp 2**: $\Delta = -0.239$, 95% CI $[-0.292, -0.185]$, $p_{\text{Holm}} < 0.001$, $W = 3.0$; Line wins 23/25). In the exploratory contrast with headers (**Comp 5**), line windows remain significantly superior ($\Delta = -0.134$, 95% CI $[-0.199, -0.070]$, $p_{\text{raw}} = 0.0003$, $W = 36.0$; Line wins 20 vs. AST wins 5).

C. Subgroups, Morphology Sensitivity, and Header Effects

Subgroups & Common Support: Trends hold across languages: Java ($N = 15$) yields Line+H 0.716 vs. AST+H 0.583 ($\Delta = -0.133$); React/TS ($N = 10$) yields 0.478 vs. 0.342 ($\Delta = -0.136$). AST fallback is 0.0% (0/438 chunks, 0/250 top-10 candidates). Evaluating on common support (826/882 pairs where both strategies have candidate chunks, retaining 91.8% of relevant evidence across all 25 queries), AST+H remains significantly below Line+H (0.489 vs. 0.600, $\Delta = -0.111$, $p_{\text{Holm}} = 0.0036$, $W = 51.0$), and AST No-H remains below Line No-H (0.342 vs. 0.535, $\Delta = -0.193$, $p_{\text{Holm}} < 0.001$, $W = 11.0$). Restricting to common support removes only 17.3% of the end-to-end gap (the gap shrinks by 0.023, $p = 0.0015$), indicating that most of the gap (82.7%) is not explained by coverage.

Window Overlap Sensitivity: Suppressing intersecting overlapping windows from the same file in the top-10 (26.4% of Line top-10 slots involved in overlap [16.8% redundant], vs. 0% in AST No-Header and 0.8% in AST+Header, the latter due to nested units; Section III-B) lowers Line NDCG@10 slightly (Line+H 0.621 to 0.583, Line No-H 0.580 to 0.574) but preserves the ranking of conditions: Line+H 0.583, Line No-H 0.574, AST+H 0.482, AST No-H 0.341. Line windows retain their lead (**Comp 5**: $\Delta = -0.101$, unadjusted $p = 0.0013$; **Comp 1**: $\Delta = -0.092$, unadjusted $p = 0.0236$), and the AST header boost remains large (**Comp 3**: $\Delta = +0.141$, unadjusted $p < 0.001$).

Qualitative Inspection of Header Lift: Inspecting rank trajectories illustrates how headers ground isolated chunks. For `JD_JAVA_01` (*"Spring Security & JWT"*; JWT = JSON Web Token), a chunk in `AccountController.java` (lines 39–64, `me()`, AST arm) that retrieves authenticated user context via `userService.getUser()` rises from Rank 6 without header to Rank 1 with header (human relevance 2). For `JD_REACT_01` (*"React Admin Dashboard & RBAC"*; RBAC = role-based access control), an administrative control panel in `Admin/index.tsx` (lines 544–613, AST arm) jumps from Rank 13 to Rank 1 with header (relevance 2).

V. DISCUSSION AND THREATS TO VALIDITY

A. Underlying Mechanisms and Scope

Node-level AST extraction (without sibling merging as in cAST [3]) does not inherently surpass sliding windows in requirement evidence retrieval. Three mechanisms may explain this; we did not test them directly: (1) *Information density and token budget*: 50-line windows (mean 41.0 LOC, 390.8 tokens) capture surrounding declarations, annotations, and helper calls, providing richer co-occurrence signals than isolated AST nodes (mean 26.7 LOC, 250.8 tokens). This token gap is intrinsic to single-node extraction; without size-budgeted sibling merging to coalesce related nodes, fine-grained AST slicing sacrifices contextual density. (2) *Contextual compensation*: headers may inject missing architectural namespace, which may allow cross-encoders to match requirement keywords against file and class names. Notably, AST headers carry more metadata (class and method cues) than line headers, yet line windows remain superior. (3) *Query granularity*: composite JD queries (5–8 skills) may favor broader 50-LOC windows encompassing co-occurring functions over individual nodes.

B. Threats to Validity

Internal Validity: Mitigated by 100% pooling at top-5 across all 12 systems (100% Judged@5, zero orphaned pairs) and blinded dual annotation ($\kappa_w = 0.9822$). Confounds from AST line coverage (69.0% lines across 183/189 files; median 16 vs. 50 LOC; 15.8% vs. 28.6% exceeding 512 tokens), header metadata asymmetry, and window overlap (26.4% of Line top-10 slots involved in overlap [16.8% redundant], vs. 0% in AST No-Header and 0.8% in AST+Header due to nested units) were audited: common-support evaluation ($N = 826$ shared pairs) shows Line retains its lead ($\Delta = -0.111, p_{\text{Holm}} = 0.0036$), and deduplication preserves the ranking. The reranker truncates inputs at 512 tokens; 28.6% of line windows and 15.8% of AST chunks exceed this length before query prepending, so tail code of long chunks may not be scored. Pooling depth (5) is smaller than evaluation depth (10); unjudged items at ranks 6–10 receive zero gain, which may bias NDCG@10 downward (bpref yields identical condition rankings). The two headers carry different amounts of information (File and Lines for line windows; File, Class, Method, and Lines for AST chunks), so the header effect is confounded with header richness. The Tree-sitter grammars were extended before the final pooling and annotation round; no relevance judgments were collected with an earlier parser configuration. **Statistical Conclusion Validity:** $N = 25$ queries provide 80% power to detect paired differences in the minimum detectable effect (MDE) range of $\Delta \in [0.08, 0.13]$ across

TABLE III
STRUCTURED RESEARCH ROADMAP AND MILESTONES

Phase	Objective	Target Data	Analysis Method
P1	Corpus Expansion (RQ3)	100 repos (4 langs); 50 JDs/issues	Factorial; mixed-effects models
P2	Graph Headers (RQ3)	Call-graphs & imports vs. path headers	Paired Wilcoxon-Holm; ablations
P3	Human Study (RQ4)	$N = 20$ software developers	Accuracy, inspection time, confidence

observed standard deviations ($\sigma_d \in [0.137, 0.184]$ across $\alpha \in [0.0125, 0.05]$); non-significant small differences (e.g., Comp 4, $\Delta = +0.041$) indicate absence of evidence rather than equivalence. **Construct Validity:** Requirements were extracted from job postings via `gemini-1.5-flash` to reflect authentic industry expectations. Relevance was judged on a 3-point ordinal graded-relevance scale with task-specific definitions (Section III-C). **External Validity:** Evaluated on Java Spring Boot and React/TypeScript with BGE-M3 and BGE-Reranker-Base; caution is warranted when generalizing to dynamically typed languages or distributed architectures.

VI. RESEARCH PLAN AND ROADMAP

Table III delineates planned phases and mitigations for RQ3 and RQ4: corpus expansion across four languages (**P1**), graph-enriched headers via precomputed SCIP/LSIF code-index formats (**P2**), and Latin-square developer verification studies (**P3**).

VII. CONCLUSION

In this ERA paper, we presented a controlled 2×2 factorial evaluation crossing chunking strategy with context headers across 40 repositories. At reranking, line windows with headers achieved the highest NDCG@10 (0.621), significantly outperforming AST chunks with headers ($\Delta = -0.134$, unadjusted $p = 0.0003$). While context headers provide a substantial boost for AST chunks ($\Delta = +0.146$, $p_{\text{Holm}} < 0.001$), line windows without headers significantly outperform AST chunks with headers ($\Delta = -0.093$, $p_{\text{Holm}} = 0.0438$) and without headers ($\Delta = -0.239$, $p_{\text{Holm}} < 0.001$). With the final parser configuration, the AST fallback rate was 0.0%. Together with the coverage and overlap audits, this indicates that, in our setting, node-level AST extraction without sibling merging does not surpass sliding windows in requirement evidence retrieval. We pose three open questions for future investigation: (1) *Can graph-enriched headers bridge the information gap for fine-grained AST chunks?* (2) *How should code-RAG benchmarks weigh syntactic integrity against contextual information density?* (3) *How can requirement evidence retrieval best assist practitioners in verifying codebase capabilities?*

Data Availability: The replication package is available at: https://anonymous.4open.science/r/ast-rag-replication-2027-82C5/.

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